

Appendix: SHRXL-200102AX

This document is applicable to RB-VOGUI XL SHRXL-200102AA and SHRXL-200102AB.

This robot has several differences with the standard RB-VOGUI XL. Nevertheless, the RB-VOGUI XL manuals are applicable to the robot, taking into account the considerations listed in this document.

For further information, please contact Robotnik support.

1 Safety - Specific risk situations



Do not approach the robot to the charging station with the charging being activated. This can lead to a small electric arc when the robot gets in contact with the station.

Always approach the robot with the charging contacts deactivated (circle button).



Do not remove the robot from the charging station with the charging activated. Always disconnect the charging (circle button) before moving the robot away.



Always maintain a safe distance to the robot when controlling it with the gamepad.



Maintain a safe distance to the robot and the charging station when approaching the robot to the station.

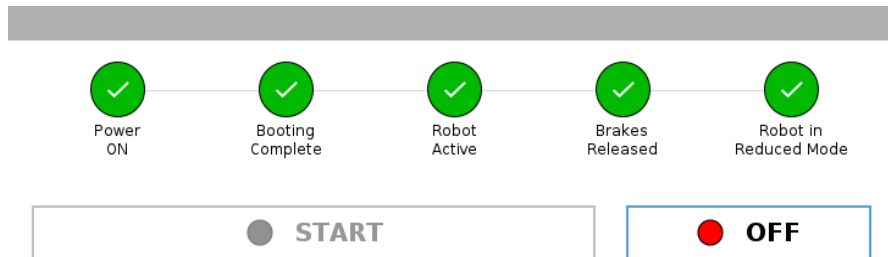


Avoid touching the robot charging contacts with a tool or a metallic element when the robot is switched on. Otherwise, in case the charging has remained activated by mistake, you may cause a short circuit that can damage the robot or injure

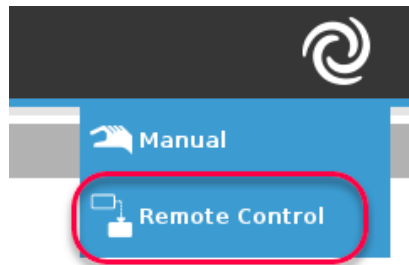
2 UR startup - Teach Pendant & ROS 2

2.1 Setup

Start the UR arm using the Teach Pendant.



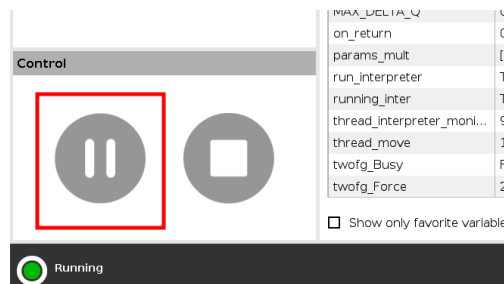
Make sure that remote control is enabled.



Make sure the External Control is loaded as a program. Then, execute the program from ROS using the following service.

```
ros2 service call /robot/arm/dashboard_client/play std_srvs/srv/Trigger {}
```

Make sure that the program is running.



Make sure that *joint_trajectory_controller* is active.

```
ros2 control list_controllers --ros-args -r __ns:=/robot/arm
```

```
robot@shrxl-200102aa:~$ ros2 control list_controllers --ros-args -r __ns:=/robot/arm
[INFO] [1768405445.143173364] [robot.arm._ros2cli_775844]: waiting for service /robot/arm/controller_manager
become available...
joint_trajectory_controller      joint_trajectory_controller/JointTrajectoryController      active
forward_velocity_controller      velocity_controllers/JointGroupVelocityController          inactive
forward_position_controller      position_controllers/JointGroupPositionController          inactive
scaled_joint_trajectory_controller ur_controllers/ScaledJointTrajectoryController             inactive
passthrough_trajectory_controller ur_controllers/PassthroughTrajectoryController             inactive
force_mode_controller           ur_controllers/ForceModeController                         inactive
tcp_pose_broadcaster            pose_broadcaster/PoseBroadcaster                           active
```

If not, active it:

```
ros2 control switch_controllers --activate joint_trajectory_controller
--ros-args -r __ns:=/robot/arm
```

```
robot@shrxl-200102aa:~$ ros2 control switch_controllers --activate joint_trajectory_controller
[INFO] [1768405379.600356556] [robot.arm._ros2cli_764045]: waiting for service /robot/arm/controller_manager
become available...
Successfully switched controllers!
Activated controllers: [ joint_trajectory_controller ]
```

2.2 Control

Run MoveIt to control the arm from ROS 2.

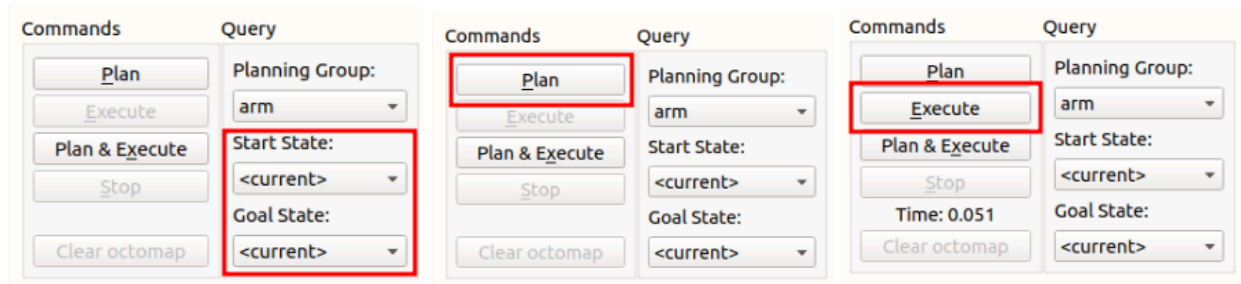
For SHRXL-200102AA

```
ros2 launch robot_bringup robot_moveit.launch.py
moveit_package:=shrxl-200102aa_moveit_config
```

For SHRXL-200102AB:

```
ros2 launch robot_bringup robot_moveit.launch.py
moveit_package:=shrxl-200102ab_moveit_config
```

It is highly recommended that, the first time MoveIt is launched, you plan and execute using the same start and goal state to account for small mechanical differences between the hardware and MoveIt.



Use the Interactive Market to move the robot into a valid position.



2.3 Troubleshooting

Arm trajectory abort immediately

1. Make sure that program is running on Polyscope. Use play service to run it again.
2. Make sure that *joint_trajectory_controller* is active.

Arm trajectory stops

1. Do not run GUI interfaces, especially RViz 2 or MoveIt, on the same machine where the control is executed, as this may degrade performance and cause trajectory execution to stop.

3. Charging Station calibration

Before performing the charging procedure, calibrate the charging station.

Although standard charging stations share the same calibration values, the marker position differs on these robots.

Measure the distance between the marker and the charging contacts, as shown in the image below:



Inside of the robot, open and edit the following file.

```
nano
~/robot_ws/src/robot_packages/packages/robot_bringup/launch/behavior/charge
.launch.py
```

Replace the default value with the measured distance.

```
None
if robot_model in ['rb1', 'rbtheron']:
    params['charge_contact_distance_from_marker'] = 0.13
else:
    params['charge_contact_distance_from_marker'] = <distance>
```

Restart the software to apply changes:

```
~/bringup.sh
```